DISCRETENESS, FINITE RANK, AND STABILITY OF P-FRAME ENERGY MINIMIZERS

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ABSTRACT. We study probability measures minimizing the energy with kernel $|x \cdot y|^p$ on the real unit sphere. For every positive rational $p = a/b$, each minimizer admits a finitely supported minimizing replacement. If a/b is reduced and $b \geq d$, every minimizer on $\mathbb{S}^{d-1}$ is finitely supported. The latter assertion follows from a finite-rank algebraic lift, Nash curve selection, and a Wronskian multiplicity bound. For irrational exponents, we prove that a positive semidefinite restriction of finite rank cannot have infinite compact support. We also establish local stability and a quantitative support bound under positive-definite Hessians at every contact of every limiting minimizer. This hypothesis is stronger than finite support and does not prove unconditional openness of the discreteness regime. Finally, subtracting a single suitably scaled high even power produces a uniformly nearby kernel whose every minimizer has finite support. We identify the atomic-purity selection of these approximants and prove a target-centered recovery theorem, separating finite approximation from discreteness of the limiting problem.

1. INTRODUCTION

For $d \geq 2$ and $p > 0$, consider the energy

$$I_p(\mu) = \iint_{\mathbb{S}^{d-1} \times \mathbb{S}^{d-1}} |x \cdot y|^p \, d\mu(x) \, d\mu(y), \qquad \mu \in \mathcal{P}(\mathbb{S}^{d-1}), \tag{1.1}$$

where $\mathcal{P}(\mathbb{S}^{d-1})$ denotes the Borel probability measures on the real unit sphere. This problem links the geometry of finite frames with the structure of equilibrium measures. Its central difficulty is not the existence of a minimizer, which follows from compactness, but whether minimizing mass must concentrate on finitely many directions.

Even exponents exhibit a basic obstruction to discreteness. When $p = 2k$, the kernel is polynomial and the energy depends on a finite tensor moment; the rotation-invariant probability measure is a minimizer. By contrast, for $0 < p < 2$, every minimizing measure is carried by the lines of an orthonormal basis [7]. The conjecture of Bilyk, Glazyrin, Matzke, Park, and Vlasiuk asks whether every minimizer is finitely supported whenever p is not an even integer [2, Conjecture 1.1]. Their work proves that such minimizing supports have empty interior. Further exact minimizers and classification results in tight-design regimes appear in [3]. Empty interior, however, leaves substantial room for infinite supports.

A useful starting point is that the kernel need not be positive semidefinite on the whole sphere, but its restriction to a minimizing support is positive semidefinite. We exploit this restriction in two ways. Algebraic powers constrain its rank and the geometry of its support. Independently, strict curvature of the minimizing potential isolates contact points and makes finiteness stable under perturbation. The hypotheses and conclusions of these mechanisms are different and must not be conflated.

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Write

$$e(p) = \min_{\mu \in \mathcal{P}(\mathbb{S}^{d-1})} I_p(\mu), \qquad \mathcal{M}_p = \{\mu : I_p(\mu) = e(p)\}, \qquad D_m = \binom{d+m-1}{m}.$$

The rank of a kernel is the supremum of the ranks of its finite Gram matrices. Our first result concerns all positive rational exponents, including the even integers in its finite-replacement assertion.

Theorem 1.1 (Rational exponents)**.** *Let $p = a/b > 0$, with $a, b \in \mathbb{N}$ coprime.*

(1) *For every $\mu \in \mathcal{M}_p$, there exists $\nu \in \mathcal{M}_p$ with $\operatorname{supp}\nu \subseteq \operatorname{supp}\mu$ and*

$$\#\operatorname{supp}\nu \le R_{d,a}, \qquad R_{d,a} = \begin{cases} D_a, & a \text{ even}, \\ D_{2a}, & a \text{ odd}. \end{cases} \tag{1.2}$$

The potentials of μ and ν agree on $\operatorname{supp}\mu$.

(2) *If an infinite compact set $S \subseteq \mathbb{S}^{d-1}$ carries a positive semidefinite restriction of $|x \cdot y|^{a/b}$, then $b < d$. Consequently, if $b \ge d$, every member of $\mathcal{M}_p$ has finite support.*

The second assertion is not a consequence of finite replacement alone. Its proof passes from an arbitrary compact support to an algebraic lift; a Nash arc in the lift yields meromorphic functions on a compact curve, and their vanishing orders force the denominator bound. In particular,

$$\mathcal{E}_d = \{p > 0 : p \notin 2\mathbb{N}, \text{ every } \mu \in \mathcal{M}_p \text{ has finite support}\} \tag{1.3}$$

is dense in $(0, \infty)$. This density does not establish openness or settle any prescribed irrational exponent.

Theorem 1.2 (Irrational finite-rank rigidity)**.** *Let $p > 0$ be irrational and $S \subseteq \mathbb{S}^{d-1}$ compact. If $|x \cdot y|^p$ is positive semidefinite and has finite rank on $S \times S$, then S is finite. Thus for $\mu \in \mathcal{M}_p$, finite support is equivalent to finite rank of its support kernel.*

The proof uses analytic curve selection in the local analytic closure of S, not an unjustified assertion that S itself contains a curve. The analytic obstruction is a finite-sum theorem of Cheng, Di Scala, and Yuan [5]. No finite-rank bound is asserted for an arbitrary irrational-exponent minimizer. We also give elementary, uniform bounds on intersections of minimizing supports with great circles.

For stability, define the potential and its whole contact set by

$$F_{p,\mu}(x) = \int |x \cdot y|^p \, d\mu(y), \qquad C_p(\mu) = \{x \in \mathbb{S}^{d-1} : F_{p,\mu}(x) = e(p)\}, \quad \mu \in \mathcal{M}_p.$$

The support is contained in $C_p(\mu)$, but equality is not assumed.

Theorem 1.3 (Nondegenerate-contact stability)**.** *Fix $p_0 > 2$. Suppose that, for every $\nu \in \mathcal{M}_{p_0}$ and every $x \in C_{p_0}(\nu)$, the spherical Hessian $\operatorname{Hess} F_{p_0,\nu}(x)$ is positive definite. Then there exist $\delta, h > 0$ and $N < \infty$ such that every $\mu \in \mathcal{M}_p$ with $|p - p_0| < \delta$ satisfies*

$$\operatorname{Hess} F_{p,\mu}(x) \succeq h \operatorname{Id} \quad (x \in C_p(\mu)), \qquad \#\operatorname{supp}\mu \le N.$$

Section 5 makes the bound quantitative. Finiteness of a minimizer does not imply the Hessian hypothesis. In particular, an isolated contact may be degenerate, and a centered atom split has no quadratic signed self-interaction. The theorem is therefore a conditional stability result, not a proof that the set (1.3) is open.

A different way to force finiteness is to flatten the kernel at the diagonal by a small polynomial correction.

Theorem 1.4 (Finite-support perturbations)**.** *For every $p > 0$ and every even integer $q > p$, every probability measure minimizing the continuous kernel*

$$L_{p,q}(x, y) = |x \cdot y|^p - \frac{p}{q}|x \cdot y|^q \tag{1.4}$$

has finite support. Moreover, $\|L_{p,q} - K_p\|_\infty = p/q$, where $K_p(x, y) = |x \cdot y|^p$.

We identify the limits of these corrected minimizers by a maximal projective atomic-purity principle. A rotationally invariant counterexample shows why finite support for every corrected problem need not pass to the limit. Finally, a target-centered positive penalty recovers any prescribed minimizer from nearby kernels. This separates three questions: existence of a finite replacement, finiteness of every minimizer, and a uniform support bound along an approximation. The general non-even-exponent discreteness conjecture remains unresolved by these results.

Throughout, weak convergence of measures is denoted by $\rightharpoonup$. Because the kernel is even in each variable, its energy depends only on the pushforward to $\mathbb{RP}^{d-1}$. A measure on the sphere has finite support if and only if this pushforward does. Cardinality bounds on the sphere count both signs; when projective points are counted instead, this is stated explicitly.

2. Variational structure

For a continuous symmetric kernel K on a compact metric space X, we use $I_K(\eta, \xi) = \iint K \, d\eta \, d\xi$ for finite signed measures and write $I_K(\eta) = I_K(\eta, \eta)$. Its energy is continuous on $\mathcal{P}(X)$, so it attains a minimum. The following formulation records precisely which positivity follows from minimality.

Lemma 2.1 (Contact and centered positivity)**.** *Let μ minimize I_K, let $E = I_K(\mu)$, and let $S = \operatorname{supp}\mu$. Then $F_\mu(x) = \int K(x, y)\, d\mu(y) \geq E$ on X, with equality on S. For any $x_1, \ldots, x_n \in S$ and $c_1, \ldots, c_n \in \mathbb{R}$,*

$$\sum_{i,j=1}^{n} c_i c_j K(x_i, x_j) \geq E \left(\sum_{i=1}^{n} c_i \right)^2 . \tag{2.1}$$

Thus $K - E$ is positive semidefinite on S. If $E \geq 0$, so is K.

Proof. The right derivative of $I_K((1 - t)\mu + t\delta_x)$ at $t = 0$ gives $F_\mu(x) \geq E$. Since $\int F_\mu \, d\mu = E$, continuity gives equality on the support.

To justify signed variations even when μ is nonatomic, let $\mu_{i,\varepsilon}$ be its normalized restriction to $B(x_i, \varepsilon)$ and set

$$s = \sum_i c_i, \qquad \eta_\varepsilon = \sum_i c_i \mu_{i,\varepsilon} - s\mu.$$

This measure has total mass zero and density

$$g_\varepsilon = \frac{d\eta_\varepsilon}{d\mu} = \sum_i \frac{c_i \mathbf{1}_{B(x_i,\varepsilon)}}{\mu(B(x_i, \varepsilon))} - s.$$

All denominators are positive because $x_i \in S$, and g_ε is bounded for fixed ε. Thus $\mu + t\eta_\varepsilon = (1 + tg_\varepsilon)\mu$ is a probability measure for both signs of t with $|t|\|g_\varepsilon\|_\infty < 1$. Also $I_K(\mu, \eta_\varepsilon) = 0$, since $F_\mu = E$ on S. Minimality therefore gives $I_K(\eta_\varepsilon) \geq 0$. Expanding yields

$$\sum_{i,j} c_i c_j I_K(\mu_{i,\varepsilon}, \mu_{j,\varepsilon}) \geq Es^2.$$

Now $\mu_{i,\varepsilon} \rightharpoonup \delta_{x_i}$; continuity of the kernel proves (2.1). □

For K_p, the minimum is strictly positive. Indeed, if $M_\mu = \int xx^{\mathsf{T}}\, d\mu(x)$, then

$$I_2(\mu) = \operatorname{tr}(M_\mu^2) \geq \frac{1}{d}, \qquad \operatorname{tr} M_\mu = 1.$$

Comparing $|t|^p$ with t^2 when $p \leq 2$, and applying Jensen's inequality to $|x \cdot y|^2$ when $p \geq 2$, yields

$$e(p) \geq \begin{cases} d^{-1}, & 0 < p \leq 2, \\ d^{-p/2}, & p \geq 2. \end{cases} \tag{2.2}$$

In particular, both $K_p - e(p)$ and K_p are positive semidefinite on any minimizing support. Such a support spans $\mathbb{R}^d$: otherwise a unit vector orthogonal to its span would have zero potential, contradicting (2.2) and the contact inequality.

Proposition 2.2 (The range $0 < p < 2$; cf. [7])**.** *If $0 < p < 2$, every minimizing measure is supported on the d lines of an orthonormal basis, each line having total mass $1/d$. Conversely every such measure minimizes. Its mass may be split arbitrarily between the two unit vectors on each line.*

Proof. A basis measure attains $1/d$, so equality must hold in $I_p(\mu) \geq I_2(\mu) \geq 1/d$. Equality in the first inequality implies $|x \cdot y| \in \{0, 1\}$ for every support pair: the continuous nonnegative difference of the two kernels has zero integral. Distinct support lines are therefore orthogonal. Equality in the second inequality is $M_\mu = d^{-1}\mathrm{Id}$, which forces exactly d such lines with equal masses. The converse is immediate. □

For comparison, if $p = 2k$ and σ is rotation-invariant probability, then σ minimizes. To see this, write

$$I_{2k}(\mu) = \left\| \int x^{\otimes 2k}\, d\mu(x) \right\|^2.$$

Averaging the tensor over the orthogonal group is an orthogonal projection and sends every such moment to the moment of σ. This proves $I_{2k}(\mu) \geq I_{2k}(\sigma)$ and explains why even exponents must be excluded from an every-minimizer discreteness conjecture.

The contact inequality also gives information away from the existing support.

Proposition 2.3 (A global variational bound)**.** *Suppose in addition that $K(x, x) = 1$. Let μ minimize I_K, and put $E = I_K(\mu)$ and $g = F_\mu - E$. If $f \in L^\infty(\mu)$ is real, $\int f\, d\mu = 0$, and $\eta = f\mu$, then*

$$|F_\eta(z)| \leq \|f\|_\infty g(z) + \sqrt{(1 - E) I_K(\eta)} \qquad (z \in X). \tag{2.3}$$

Here $I_K(\eta) \geq 0$ and $E \leq 1$.

Proof. The asserted signs follow from bounded-density variations and comparison with a point mass. If $M = \|f\|_\infty = 0$, there is nothing to prove. Otherwise set $Q = I_K(\eta)$, $b = F_\eta(z)$, and $\rho = (1 - \operatorname{sgn}(b) f/M)\mu$, using either sign when $b = 0$. Then ρ is a probability measure,

$$I_K(\rho) = E + Q/M^2, \qquad I_K(\rho, \delta_z) = E + g(z) - |b|/M.$$

Comparison of μ with $(1 - t)\rho + t\delta_z$, followed by division by $(1 - t)^2$ and substitution $r = t/(1 - t)$, gives

$$0 \leq Q/M^2 + 2r\big(g(z) - |b|/M\big) + r^2(1 - E) \qquad (r \geq 0).$$

For $a, d \geq 0$, a quadratic $a + 2cr + dr^2$ is nonnegative on $[0, \infty)$ only if $c \geq -\sqrt{ad}$. This proves (2.3), including the cases $Q = 0$ or $E = 1$. □

For example, if $I_K(\eta) = 0$, then F_η vanishes on the whole contact set, not only on the support. This is a restriction on an existing exact null direction; it does not prove that such a direction exists on an infinite support.

3. Rational exponents: rank and the denominator obstruction

3.1. **Finite minimizing replacements.**

Lemma 3.1 (Rank under Hadamard powers)**.** *If H is a positive semidefinite kernel and $m \in \mathbb{N}$, then* $\operatorname{rank} H \le \operatorname{rank}(H^{\circ m})$*, where* $H^{\circ m}(x,y) = H(x,y)^m$*.*

Proof. Represent $H(x,y) = \langle h_x, h_y \rangle$ in a real Hilbert space. The kernel $H^{\circ m}$ has feature vectors $h_x^{\otimes m}$. If $h_{x_1}, \dots, h_{x_r}$ are linearly independent, choose linear functionals ℓ_i on their span with $\ell_i(h_{x_j}) = \delta_{ij}$. Applying $\ell_i^{\otimes m}$ shows that their tensor powers are also independent. Taking the supremum over finite independent families proves the assertion, including infinite rank. □

For $p = a/b$ in lowest terms,

$$K_p^{\circ 2b}(x,y) = (x \cdot y)^{2a}. \tag{3.1}$$

The polynomial kernel on the right has feature map $x^{\otimes 2a}$ in $\operatorname{Sym}^{2a} \mathbb{R}^d$, of dimension D_{2a}. When a is even one may instead use $K_p^{\circ b} = (x \cdot y)^a$. Thus every positive semidefinite restriction of K_p has rank at most $R_{d,a}$ in (1.2).

Proof of Theorem 1.1(1). Let $S = \operatorname{supp} \mu$. Lemma 2.1 and (2.2) give a continuous finite-rank Gram representation

$$K_p(x,y) = \langle \Phi(x), \Phi(y) \rangle, \qquad x, y \in S, \qquad \Phi : S \longrightarrow \mathbb{R}^R,$$

where $R \le R_{d,a}$. For completeness, choose R anchors with invertible Gram matrix G and take $\Phi(x) = G^{-1/2}(K_p(x, x_i))_{i=1}^R$; the rank condition gives the displayed identity.

Set $M = \int \Phi \, d\mu$. Then $\|M\|^2 = e(p) > 0$ and $\langle \Phi(x), M \rangle = e(p)$ for $x \in S$. Hence $\Phi(S)$ lies in a proper affine hyperplane of $\mathbb{R}^R$. Its compact convex hull contains M, so affine Carathéodory gives $M = \sum_{i=1}^n w_i \Phi(x_i)$ with $n \le R$, $x_i \in S$, $w_i \ge 0$, and $\sum_i w_i = 1$. The measure $\nu = \sum_i w_i \delta_{x_i}$ has $I_p(\nu) = \|M\|^2 = e(p)$ and, for every $x \in S$, $F_{p,\nu}(x) = \langle \Phi(x), M \rangle = F_{p,\mu}(x)$. □

3.2. **A multiplicity lemma on a compact curve.**

We use two standard facts about algebraic curves. A nonconstant Nash arc has coordinate functions algebraic over the parameter field, so its complex Zariski closure is an irreducible curve. Normalizing the projective closure gives a compact connected Riemann surface on which the coordinate functions are meromorphic. Nash curve selection is available for a nonisolated point of a real algebraic set [4, Proposition 8.1.13]; normalization and meromorphic divisor theory are standard [8].

Lemma 3.2 (Finite-dimensional vanishing orders)**.** *Let V be an r-dimensional complex vector space of meromorphic functions on a compact connected Riemann surface X. There is a finite set $T \subset X$ such that every nonzero $f \in V$ has vanishing order at most $r-1$ at every point of $X \setminus T$. The poles of all functions in V are bounded by a common effective divisor.*

Proof. Choose a basis $f_1, \dots, f_r$ and form its Wronskian in a local coordinate. It is not identically zero: at a point without poles, linear elimination gives a basis of germs with distinct initial orders; the leading Wronskian coefficient is a nonzero Vandermonde determinant in those orders. Under coordinate changes the Wronskian is multiplied by a nonvanishing factor, and thus defines a meromorphic section of a tensor power of the canonical bundle, with finitely many zeros and poles. Include these points and the poles of the f_i in T. Off T, the map assigning the first r Taylor coefficients to a member of V is invertible, proving the order bound. A common pole divisor is

$$D = \sum_{z \in X} \max_i \{0, -\operatorname{ord}_z f_i\} \, [z].$$

□

3.3. **From arbitrary compact support to a curve.**

Proof of Theorem 1.1(2). Suppose S is infinite and compact and $K_{a/b}|_{S\times S}$ is positive semidefinite. By Lemma 3.1, there is a continuous finite-dimensional feature map $h : S \to \mathbb{R}^R$ with

$$\langle h(x), h(y)\rangle = |x \cdot y|^{a/b}, \qquad \|h(x)\| = 1.$$

Let $A = \{(x, h(x)) : x \in S\}$, and let Z be its real Zariski closure, that is, the common real zero set of all polynomials vanishing on A. By the Hilbert basis theorem this is a real algebraic set. Since $\|x\|^2 = \|u\|^2 = 1$ on A, these equations hold on Z, so Z is compact. The identity

$$(u \cdot v)^{2b} = (x \cdot y)^{2a} \tag{3.2}$$

holds for all $(x, u), (y, v) \in Z$: first fix one argument in A and extend polynomially in the other, then extend in the first argument. If $(x, u), (x, v) \in Z$, the unit norm equations and (3.2) give $v = \pm u$. Thus projection onto the x-coordinates has fibers of size at most two.

The infinite compact algebraic set Z has a nonisolated point z_0. Apply Nash curve selection to the semialgebraic set $Z \setminus \{z_0\}$ at z_0. It gives a nonconstant Nash arc $\gamma : [0, \varepsilon) \to Z$, with $\gamma(0) = z_0$ and $\gamma(t) \neq z_0$ for $t > 0$. Write $\gamma(t) = (x(t), u(t))$. Its projection $x(t)$ is nonconstant, since a continuous arc in a finite fiber is constant.

Evaluation on the complexified germ of γ maps the complex polynomial ring into the integral domain $\mathbb{C}\{t\}$, so its kernel is prime. Since the coordinate germs are Nash, $\mathbb{C}(t, \gamma_1, \ldots, \gamma_{d+R})$ is algebraic over $\mathbb{C}(t)$. Its subfield generated by the coordinate germs has transcendence degree at most one, and nonconstancy makes that degree one. The complex Zariski closure C of the arc is therefore an irreducible affine curve. The arc is Zariski dense in C, and successive extension in each variable makes $\gamma \times \gamma$ Zariski dense in $C \times C$. Thus the norm equations and (3.2) hold on this curve and its product. On the compact normalization X of the projective closure of C, the coordinates x_i, u_j are meromorphic and the same identities hold meromorphically on X and $X \times X$.

Let W be the real span of the projected real arc and let $r = \dim W \leq d$. In a real orthonormal basis of W, write its meromorphic coordinates as $f_1, \ldots, f_r$. They are complex-linearly independent: a complex linear relation restricts to two real relations on a spanning real arc. Moreover,

$$\sum_{i=1}^{r} f_i^2 = 1.$$

For a regular parameter $t \in X$, define meromorphic functions of s by

$$A_t(s) = \sum_{i=1}^{r} f_i(s) f_i(t), \qquad B_t(s) = \sum_{j=1}^{R} u_j(s) u_j(t).$$

They satisfy

$$B_t^{2b} = A_t^{2a}, \qquad A_t(t) = B_t(t) = 1. \tag{3.3}$$

Since a, b are coprime, taking orders in (3.3) gives

$$b \mid \operatorname{ord}_z A_t \qquad (z \in X). \tag{3.4}$$

Apply Lemma 3.2 to $V = \operatorname{span}_{\mathbb{C}}\{f_1, \ldots, f_r\}$. If $b \geq r$, then (3.4) forces every zero of every A_t into its fixed finite exceptional set T. Their poles are bounded by the fixed divisor D of that lemma. For every t,

$$\deg(A_t)_0 = \deg(A_t)_\infty \leq \deg D.$$

Thus zero multiplicities at the finitely many points of T are uniformly bounded as well. Only finitely many principal divisors (A_t) can occur. Functions with the same divisor differ by a nonzero scalar, so the projective classes $[A_t] \in \mathbb{P}(V)$ take only finitely many values.

On a connected open set where all coordinates are regular, the map $t \mapsto [A_t]$ is holomorphic and hence must be constant. Independence of the f_i would then give $f_i(t) = c_i g(t)$ for fixed constants

c_i and one holomorphic function g. The identity $\sum_i f_i(t)^2 = 1$ forces $g^2 \sum_i c_i^2 = 1$, so g and all the f_i are constant. Analytic continuation contradicts nonconstancy of $x(t)$. Thus $b < r \le d$. Applying Lemma 2.1 to a minimizing support proves the last assertion of the theorem. □

Corollary 3.3. *The set $\mathcal{E}_d$ in (1.3) is dense in $(0, \infty)$.*

Proof. It contains every positive rational with reduced denominator at least d. Any bounded interval contains infinitely many rationals but only finitely many with reduced denominator less than d. □

The denominator obstruction is sharp for positive semidefinite supports, without a minimality assumption. Let $v(t) = (\cos t, \sin t)$ and $x(t) = v(t)^{\otimes b} \in \mathrm{Sym}^b \mathbb{R}^2 \cong \mathbb{R}^{b+1}$. On a sufficiently short closed interval,

$$x(s) \cdot x(t) = \cos^b(s-t) > 0, \qquad |x(s) \cdot x(t)|^{a/b} = \cos^a(s-t).$$

The last kernel is the Gram kernel of $v(t)^{\otimes a}$. This gives an infinite compact positive semidefinite support in dimension $d = b + 1$. It does not supply a minimizing measure for the sphere problem.

4. Analytic rigidity and great-circle restrictions

4.1. **Finite rank at an irrational exponent.** We use the following consequence of [5, Theorem 2.1(iii)]. If holomorphic germs $\chi_1, \dots, \chi_s$ vanish at 0, at least one is nonconstant, and

$$\left(1 + \sum_{i=1}^{s} |\chi_i(z)|^2\right)^{\alpha} = \sum_{j=1}^{R} |Q_j(z)|^2 \tag{4.1}$$

for finitely many holomorphic germs Q_j, then $\alpha \in \mathbb{Q}$. Indeed, the right side belongs to the finite-sum Umehara algebra, and hence to its field of fractions. The cited theorem applies after deleting identically zero χ_i; the expression on the left is nonconstant. We also use Noetherianity of the ring of real-analytic germs and the real-analytic curve-selection lemma for a semianalytic zero set; see [1]. This is distinct from the Nash curve selection used in Section 3.

Proof of Theorem 1.2. Suppose S is infinite. Choose an accumulation point x_0 and let A be the intersection of S with a sufficiently small closed spherical cap centered at x_0, so that all pairwise inner products in A are positive. Let the rank of the restricted kernel on A be r, choose r anchors $s_1, \dots, s_r \in A$ with invertible Gram matrix G, and set

$$h(x) = G^{-1/2}\big((x \cdot s_i)^p\big)_{i=1}^{r}.$$

This defines a real-analytic map in an ambient neighborhood U of x_0 where the relevant inner products are positive. Shrink U so that $x \cdot y > 0$ for $x, y \in U$ and $S \cap U \subset A$. Finite-rank Gram linear algebra gives

$$R(x, y) := (x \cdot y)^p - h(x) \cdot h(y) = 0 \qquad (x, y \in S \cap U).$$

Let $\mathcal{I}$ be the ideal of real-analytic germs at x_0 vanishing on S in some neighborhood of x_0. By Noetherianity of the convergent power-series ring $\mathbb{R}\{x - x_0\}$, choose germ generators $g_1, \dots, g_m$ and representatives on a common smaller neighborhood V. Then $S \cap V \subset Z := \{g_1 = \cdots = g_m = 0\}$, and x_0 is nonisolated in Z. Apply real-analytic curve selection to the semianalytic set $Z \setminus \{x_0\}$ at x_0. This gives a nonconstant analytic arc $\gamma(t)$ through x_0 in Z for $t \ge 0$. Its convergent series extend to a two-sided interval, and the identity theorem keeps every $g_i(\gamma(t))$ zero there. Since $\|x\|^2 - 1 \in \mathcal{I}$, the arc lies on the sphere. It is not claimed to lie in S.

For each fixed $y \in S \cap V$, the germ $R(\cdot, y)$ belongs to $\mathcal{I}$. It vanishes on $\gamma(t)$ initially near 0, and then on a common connected interval by the one-variable identity theorem. Consequently, for each

fixed t in that interval, $R(\gamma(t),\cdot)$ vanishes on $S\cap V$ and belongs to $\mathcal{I}$. Applying the same argument in the second variable gives

$$\big(\gamma(t)\cdot\gamma(u)\big)^p = h(\gamma(t))\cdot h(\gamma(u)) \tag{4.2}$$

for all sufficiently small real t,u. This two-step continuation avoids any assumption of a uniform radius for ideal membership as a parameter varies.

Rotate so that $\gamma(0)=e_1$ and extend the real-analytic functions holomorphically. The identity theorem, applied successively in the two parameters of (4.2), gives the polarized holomorphic identity near $(0,0)$, with the power branch fixed by its value at $(0,0)$. Substitution of the antiholomorphic diagonal $u=\bar z$ yields

$$\sum_{j=1}^{r}|h_j(\gamma(z))|^2=\left(\sum_{i=1}^{d}|\gamma_i(z)|^2\right)^p.$$

The germ γ_1 is nonzero near 0 and admits a holomorphic pth power there. Divide by $|\gamma_1(z)|^{2p}$ and put

$$\chi_i=\gamma_i/\gamma_1\quad(2\le i\le d),\qquad Q_j=(h_j\circ\gamma)/\gamma_1^p.$$

All the χ_i vanish at 0, and at least one is nonconstant; otherwise the real unit arc would be constant. Delete the identically zero χ_i; every remaining germ is nonconstant. We obtain (4.1) with $\alpha=p$. Its nonconstant left side equals a finite sum of squared moduli and therefore belongs to the nonconstant part of the Umehara fraction field. Theorem 2.1(iii) of [5] forces $p\in\mathbb{Q}$, a contradiction. The equivalence for minimizing measures now follows from Lemma 2.1, since a kernel on a finite set has finite rank. □

4.2. **Uniform finiteness on great circles.** A different argument bounds intersections with a great circle without assuming finite rank. The quantitative distinction between a short-arc bound and a whole-circle bound is important.

Theorem 4.1. *For every $p>0$ with $p\notin 2\mathbb{N}$, there is a finite number $B(p)$ such that every minimizing support in every dimension meets each great circle in at most $B(p)$ points. If $p=m$ is odd, one may take $B(p)=10m$. If p is noninteger, there is $\delta(p)>0$ such that one may take*

$$B(p)=(\lfloor p\rfloor+2)\left\lceil\frac{2\pi}{\delta(p)}\right\rceil.$$

In particular, every minimizer on $\mathbb{S}^1$ has finite support.

Proof. First let p be noninteger, set $a=\lfloor p\rfloor$, and put $N=a+3$. Parametrize a short great-circle arc by $x(u)=(z+uv)/(1+u^2)^{1/2}$, where z,v are orthonormal. Its kernel matrix is positively diagonally congruent to

$$G_{ij}=(1+u_iu_j)^p.$$

For distinct nodes, divided-difference elimination in rows and columns gives

$$\det G=\prod_{i<j}(u_j-u_i)^2\det H,\qquad H_{ij}=g[u_1,\dots,u_i;u_1,\dots,u_j], \tag{4.3}$$

where $g(u,v)=(1+uv)^p$. The integral formula for divided differences [6, Chapter I] gives, uniformly as all nodes tend to 0,

$$H_{ij}\longrightarrow\frac{\partial_u^{i-1}\partial_v^{j-1}g(0,0)}{(i-1)!(j-1)!}=\delta_{ij}\binom{p}{i-1}.$$

Uniformity holds regardless of how closely some nodes approach each other. The coefficients $\binom{p}{0},\dots,\binom{p}{a+1}$ are positive and $\binom{p}{a+2}<0$. Thus (4.3) is negative for N distinct nodes in a sufficiently short arc, contradicting positive semidefiniteness. Choose its angular length $\delta(p)$ uniformly by rotation. Covering the full circle by $\lceil 2\pi/\delta(p)\rceil$ such arcs gives the stated bound.

Now let $p = m$ be odd, S be a minimizing support, and $A \subset S$ a pairwise acute subset. Put $\phi(x) = x^{\otimes m}$. From (2.1), for every finite family in A,

$$\left\| \sum_i c_i \phi(x_i) \right\|^2 \geq e(m) \left(\sum_i c_i \right)^2 .$$

Therefore the functional $\ell(\sum_i c_i \phi(x_i)) = \sum_i c_i$ is well defined and bounded on the span of $\phi(A)$. Finite-dimensional Riesz representation yields one homogeneous odd polynomial P of degree m with $P = 1$ on A. On a great circle, $P - 1$ is a nonzero trigonometric polynomial of degree at most m: it cannot vanish identically because P is odd. Such a polynomial has at most $2m$ distinct zeros. Five arcs of angular length $2\pi/5 < \pi/2$ cover the circle and are pairwise acute within each arc, giving $10m$. □

This theorem controls one-dimensional sections, not the full support in higher dimensions. A latitude circle in $\mathbb{S}^2$, for example, meets each great circle in at most two points. Nor does the noninteger proof apply to an arbitrary smooth curve: its exact diagonal congruence to $(1 + uv)^p$ uses great-circle geometry. The tensor curves at the end of Section 3 already show that general analytic curves can carry positive semidefinite rational-power kernels.

5. Contact curvature and conditional stability

For $p > 2$ the potential is C^2 on the sphere. At a contact point x, its Hessian in a unit tangent direction v is

$$\operatorname{Hess} F_{p,\mu}(x)[v, v] = p(p-1) \int |x \cdot y|^{p-2} (v \cdot y)^2 \, d\mu(y) - pe(p). \tag{5.1}$$

This follows by differentiating along $x(s) = x \cos s + v \sin s$. All derivatives extend continuously across $x \cdot y = 0$ when $p > 2$. Since contacts are global minima of the potential, (5.1) is nonnegative. Strict positivity requires an additional argument or hypothesis.

5.1. **The exact centered split.** Suppose $\mu = w\delta_x + \lambda$ is minimizing, $w > 0$. Replacing its atom by $w\rho$, where ρ is a probability measure, gives the exact identity

$$I_p\big(\mu + w(\rho - \delta_x)\big) - e(p) = 2w \int (F_{p,\mu} - e(p)) \, d\rho + w^2 I_p(\rho - \delta_x). \tag{5.2}$$

Take $\rho_\varepsilon = (\delta_{x\cos\varepsilon + v\sin\varepsilon} + \delta_{x\cos\varepsilon - v\sin\varepsilon})/2$ with $v \perp x$ unit. For sufficiently small ε,

$$\begin{aligned} I_p(\rho_\varepsilon - \delta_x) &= \frac{3}{2} + \frac{1}{2}\cos^p(2\varepsilon) - 2\cos^p \varepsilon \\ &= \frac{p(3p-2)}{4}\varepsilon^4 + O(\varepsilon^6). \end{aligned} \tag{5.3}$$

The quadratic self-energy decrease cancels the contribution already present in the first-variation term. In particular, when $p > 2$, the whole variation has leading expression

$$w \operatorname{Hess} F_{p,\mu}(x)[v, v]\varepsilon^2 + o(\varepsilon^2) + w^2 \frac{p(3p-2)}{4}\varepsilon^4 + O(\varepsilon^6).$$

It cannot force strict positivity of the Hessian. Likewise, an isolated zero of a nonnegative analytic function need not be quadratic, as u^4 demonstrates.

5.2. **Stability at the whole contact set.**

Lemma 5.1. *If $p_n \to p_0 > 2$ and $\mu_n \rightharpoonup \nu$, then $F_{p_n,\mu_n} \to F_{p_0,\nu}$ in $C^2(\mathbb{S}^{d-1})$ and $I_{p_n}(\mu_n) \to I_{p_0}(\nu)$. If $\mu_n \in \mathcal{M}_{p_n}$, then $\nu \in \mathcal{M}_{p_0}$.*

Proof. On a compact parameter interval in $(2,\infty)$, the kernel and its first two spherical derivatives in the first variable are jointly continuous in (p,x,y). Uniform continuity and a finite-net argument turn weak convergence against these continuous functions of y into uniform convergence in x, also for the derivatives. Energy convergence follows by integration; comparison with each fixed competitor proves minimality of the limit. □

Proposition 5.2 (Local cardinality bound)**.** *Suppose $p_n \to p_0 > 2$, $\mu_n \in \mathcal{M}_{p_n}$, and $\mu_n \rightharpoonup \nu$. If every point of $C_{p_0}(\nu)$ has a positive-definite contact Hessian, then this contact set is finite and*

$$\#\operatorname{supp}\mu_n \le \#C_{p_0}(\nu)$$

for all sufficiently large n.

Proof. The limiting contact set is compact and every contact is isolated, so there are finitely many. Take disjoint small geodesically convex balls around them in which the limiting Hessian remains positive definite. Lemma 5.1 preserves strict convexity in these balls for large n. On their compact complement the limiting contact gap is positive; uniform convergence of potentials and minimum energies preserves that gap as well. Every support point of μ_n therefore lies in a ball, and strict geodesic convexity allows at most one global potential minimum in each ball. □

The use of *all* contacts is essential to this argument. At an unused contact $z \in C_{p_0}(\nu)$, insertion has no first-order cost:

$$I_{p_0}((1-t)\nu + t\delta_z) - e(p_0) = t^2(1 - e(p_0)). \tag{5.4}$$

Curvature only at the old support does not control vanishing mass near such a point. The evenness of the potential causes no difficulty: contacts occur in antipodal pairs, and their Hessians are isometric.

Proof of Theorem 1.3. Consider the compact set

$$\{(\nu,x,v) : \nu \in \mathcal{M}_{p_0},\ x \in C_{p_0}(\nu),\ \|v\| = 1,\ v \perp x\}.$$

The Hessian value on this set is continuous and, by hypothesis, strictly positive. It therefore has a positive minimum h_0. If no nearby uniform bound $h_0/2$ existed at all contacts of all minimizers, we could choose $p_n \to p_0$, $\mu_n \in \mathcal{M}_{p_n}$, contacts x_n, and unit tangent directions v_n violating it. Compactness and Lemma 5.1 give a limiting triple in the displayed set and a Hessian value at most $h_0/2$, a contradiction.

Choose a compact interval $J \subset (2,\infty)$ containing the resulting parameter neighborhood. Proposition 5.3 below, with the common Hessian lower bound $h_0/2$, gives a common finite support bound. □

Thus the all-contact nondegeneracy condition is open and supplies an open interval in $\mathcal{E}_d$ whenever it holds at a non-even exponent. The theorem does not identify finite support with that condition and does not establish openness of $\mathcal{E}_d$ itself.

5.3. **An explicit separation bound.**

Proposition 5.3. *Let $J = [a,b] \subset (2,\infty)$ and put*

$$\beta = \min\{1, a-2\}, \qquad C_J = b(b-1)\big(\max\{1, b-2\} + 2\big) + b^2. \tag{5.5}$$

Suppose $p \in J$, $\mu \in \mathcal{M}_p$, and $\operatorname{Hess} F_{p,\mu}(x) \succeq h \operatorname{Id}$ *for all $x \in \operatorname{supp}\mu$, where $h > 0$. Set*

$$r = \min\left\{1, \left(\frac{h}{2C_J}\right)^{1/\beta}\right\}.$$

Distinct support points have spherical distance greater than r, and

$$\#\operatorname{supp}\mu \le \left(1 + \frac{\pi}{r}\right)^d. \tag{5.6}$$

On $\mathbb{S}^2$ one also has $\#\operatorname{supp}\mu \le 2/(1-\cos(r/2))$.

Proof. For a unit-speed great circle $x(s)$, write $t(s) = x(s)\cdot y$. Then $|t|, |t'| \le 1$, $t'' = -t$, and

$$\frac{d^2}{ds^2}|t(s)|^p = p(p-1)|t(s)|^{p-2}t'(s)^2 - p|t(s)|^p.$$

On $[-1,1]$, the map $t \mapsto |t|^{p-2}$ is uniformly β-Hölder for $p \in J$, with constant at most $\max\{1, b-2\}$; for $|s| \le 1$ the same estimate follows by composing with $t(s)$. Also $|t'(s)^2 - t'(0)^2| \le 2|s|$ and $||t(s)|^p - |t(0)|^p| \le b|s|$. Integration against μ gives

$$|(F_{p,\mu}\circ x)''(s) - (F_{p,\mu}\circ x)''(0)| \le C_J|s|^\beta \qquad (|s| \le 1).$$

Starting at a support point, the first derivative vanishes and the second derivative is at least h. It remains at least $h/2$ until time r, so another contact cannot occur at any distance in $(0, r]$.

Geodesic separation greater than r gives Euclidean chordal separation greater than $2r/\pi$. The disjoint Euclidean balls of radius r/π centered at the support points lie in the ball of radius $1+r/\pi$; volume comparison proves (5.6). On $\mathbb{S}^2$, disjoint spherical caps of radius $r/2$ instead give the last bound. □

When $b \ge 3$, the constant in (5.5) simplifies to $C_J = b^3$. The estimate is conditional: no positive lower bound h for all minimizers at an arbitrary non-even exponent is supplied here.

6. Finite-support regularization at a fixed exponent

6.1. **A flat-diagonal obstruction.**

Lemma 6.1. *Let $h : [-1,1] \to \mathbb{R}$ be continuous and even, and C^2 near 1. Suppose*

$$h(1) = c > 0, \qquad h'(1) = 0, \qquad h'(t) \ne 0 \quad (1-\delta < t < 1)$$

for some $\delta > 0$. If $h(x\cdot y)$ is positive semidefinite on an infinite compact subset of $\mathbb{S}^{d-1}$, these conditions give a contradiction.

Proof. Suppose distinct support points x_n converge to x. Put $r_n = \|x_n - x\|$ and, after passing to a subsequence, assume $(x_n - x)/r_n \to u$, where $\|u\| = 1$ and $u \perp x$. Since $x_n \cdot x = 1 - r_n^2/2$, Taylor's theorem gives $c - h(x_n\cdot x) = O(r_n^4)$. A Gram representation of the positive semidefinite kernel and Cauchy–Schwarz imply, for every fixed support point y,

$$|h(x_n\cdot y) - h(x\cdot y)|^2 \le 2c\big(c - h(x_n\cdot x)\big) = O(r_n^4). \tag{6.1}$$

Now fix $y = x_j$ sufficiently close to x that $x\cdot x_j \in (1-\delta, 1)$. Divide the unsquared difference in (6.1) by r_n and let $n \to \infty$. Differentiability near this inner product gives $h'(x\cdot x_j)(u\cdot x_j) = 0$, hence $u \cdot x_j = 0$. This holds for every sufficiently large fixed j. Therefore $u\cdot(x_j - x)/r_j = 0$ for those j, and the limit $j \to \infty$ contradicts $\|u\| = 1$. □

Proof of Theorem 1.4. Let $q > p$ be even and put $h_q(t) = |t|^p - (p/q)|t|^q$. This function is nonnegative on $[-1,1]$, and its minimum energy is therefore nonnegative. By Lemma 2.1, its kernel is positive semidefinite on the support of every minimizer. Near 1,

$$h_q(1) = 1 - p/q > 0, \qquad h_q'(1) = 0, \qquad h_q'(t) = pt^{p-1}(1 - t^{q-p}) > 0 \quad (0 < t < 1).$$

Lemma 6.1 proves finiteness of each minimizing support. The uniform difference from K_p has maximum p/q, attained on the diagonal. Since q is even, the correction is a polynomial kernel of rank at most D_q. □

The correction is small in the uniform norm, but not in C^2 at the diagonal: $h_q''(1) = p(p-q)$. Thus its flat-diagonal mechanism does not persist automatically in the unmodified limiting problem.

6.2. **Which unmodified minimizers are selected?** In this subsection work on $X = \mathbb{RP}^{d-1}$, so one atom means one unoriented line. All kernels under consideration descend to X, and finite support on X is equivalent to finite support on the sphere. Define the atomic purity

$$A(\mu) = (\mu \otimes \mu)(\Delta_X) = \sum_{x \in X} \mu\{x\}^2, \qquad a_*(p) = \max_{\mu \in \mathcal{M}_p} A(\mu). \tag{6.2}$$

The maximum exists: A is upper semicontinuous by the closedness of the diagonal and the Portmanteau theorem. On the sphere, this same quantity is the sum of squares of the total masses on antipodal pairs, not the sum of squares of individual signed-point masses.

Let ν_q be any minimizer of $L_{p,q}$ and let $e_{p,q} = I_p(\nu_q) - (p/q)I_q(\nu_q)$.

Proposition 6.2 (Maximal-purity selection)**.** *As even $q \to \infty$ with p fixed, every weak cluster point of ν_q belongs to $\mathcal{M}_p$ and has purity $a_*(p)$. Moreover,*

$$I_q(\nu_q) \longrightarrow a_*(p), \tag{6.3}$$

$$I_p(\nu_q) - e(p) = o(q^{-1}), \tag{6.4}$$

$$e_{p,q} = e(p) - \frac{p}{q} a_*(p) + o(q^{-1}). \tag{6.5}$$

Proof. Write $e = e(p)$ and $a_* = a_*(p)$. Optimality against any $\mu \in \mathcal{M}_p$ gives

$$0 \le I_p(\nu_q) - e \le \frac{p}{q}\big(I_q(\nu_q) - I_q(\mu)\big) \le \frac{p}{q}. \tag{6.6}$$

In particular, all cluster points minimize I_p. Choose a purity-maximizing $\mu_* \in \mathcal{M}_p$. Since $I_q(\mu_*) \ge A(\mu_*)$, (6.6) gives $I_q(\nu_q) \ge a_*$.

Suppose $\nu_{q_j} \rightharpoonup \nu$. For every fixed $R > 0$ and large j, monotonicity of the powers on $[0,1]$ yields

$$\limsup_j I_{q_j}(\nu_{q_j}) \le \lim_j I_R(\nu_{q_j}) = I_R(\nu).$$

As $R \to \infty$, dominated convergence gives $I_R(\nu) \downarrow A(\nu)$ on projective space. Therefore

$$a_* \le \liminf_j I_{q_j}(\nu_{q_j}) \le \limsup_j I_{q_j}(\nu_{q_j}) \le A(\nu) \le a_*.$$

Compactness proves (6.3) for all choices of minimizers. Also $I_q(\mu_*) \to a_*$. Applying (6.6) to this fixed μ_* proves (6.4), and the definition of $e_{p,q}$ then gives (6.5). □

The conclusion is about $I_q(\nu_q)$, not necessarily $A(\nu_q)$. It supplies neither a positive lower bound for $a_*(p)$ nor finiteness of a limiting support. Even positive purity would only guarantee the presence of an atom, not a finite number of atoms.

6.3. **A counterexample to passage of finiteness through the limit.** The following example uses the same type of high-even-power correction, but a different rotationally invariant kernel. It isolates the failure of the approximation inference, without contradicting the p-frame conjecture.

Proposition 6.3. *On $\mathbb{RP}^{d-1}$, the kernel $B(x,y) = \exp((x\cdot y)^2 - 1)$ has a unique minimizer, namely rotation-invariant probability σ; this measure is nonatomic. For every even $q \ge 6$, every minimizer of*

$$B_q(x,y) = B(x,y) - \frac{2}{q}|x \cdot y|^q$$

has finite support. All such minimizing measures converge weakly to σ as $q \to \infty$, and their support cardinalities tend to infinity.

Proof. For any finite real signed measure η on projective space,

$$I_B(\eta) = \mathrm{e}^{-1} \sum_{k\geq 0} \frac{1}{k!} \left\| \int x^{\otimes 2k}\, d\eta(x) \right\|^2 .$$

The series is uniformly convergent at the kernel level. If its value is zero, all even polynomial moments vanish. Even polynomials separate projective points, so Stone–Weierstrass and uniqueness of measures imply $\eta = 0$. Thus the quadratic form is strictly positive on nonzero signed measures. Rotation invariance makes $F_{B,\sigma}$ constant; for a probability $\mu \neq \sigma$,

$$I_B(\mu) - I_B(\sigma) = I_B(\mu - \sigma) > 0.$$

For even $q \geq 6$, put $b_q(t) = \mathrm{e}^{t^2-1} - (2/q)t^q$. It is positive on $[-1, 1]$, since $b_q \geq \mathrm{e}^{-1} - 2/q > 0$, and $b_q(1) = 1 - 2/q > 0$. Also $b_q'(1) = 0$, while

$$b_q'(t) = 2t\big(\mathrm{e}^{t^2-1} - t^{q-2}\big) > 0 \qquad (0 < t < 1).$$

Indeed, $(q-2)\log t \leq 2\log t < t^2 - 1$ there. Lemmas 2.1 and 6.1 prove finite support of every B_q-minimizer. Uniform convergence $B_q \to B$ and uniqueness of the B-minimizer force convergence to σ. Measures with at most a fixed number of atoms form a weakly compact set, so no subsequence of these support cardinalities can remain bounded. □

7. Approximation, selection, and recovery

7.1. **Exponent perturbations and their selection rule.** On every compact positive exponent interval, the kernels are locally Lipschitz in the exponent. If $p, q \geq \alpha > 0$, then

$$\|K_p - K_q\|_\infty \leq \frac{|p-q|}{\mathrm{e}\alpha}, \tag{7.1}$$

because $\sup_{0\leq t\leq 1} t^\alpha |\log t| = 1/(\mathrm{e}\alpha)$. Thus $e(p)$ is locally Lipschitz, and every weak limit of exact minimizers at exponents tending to p lies in $\mathcal{M}_p$. This assertion neither preserves support cardinality nor says that every member of $\mathcal{M}_p$ can be recovered in that way.

To make the latter issue precise, set

$$J_p(\mu) = \iint |x \cdot y|^p \log|x \cdot y|\, d\mu(x)\, d\mu(y),$$

with the integrand defined as zero at $x \cdot y = 0$.

Proposition 7.1 (One-sided selection)**.** *If $h_n \downarrow 0$ and $\mu_n \in \mathcal{M}_{p+h_n}$, every weak limit minimizes J_p over $\mathcal{M}_p$. Limits from exponents $p - h_n$ maximize J_p over $\mathcal{M}_p$. Moreover,*

$$e'_+(p) = \min_{\mathcal{M}_p} J_p, \qquad e'_-(p) = \max_{\mathcal{M}_p} J_p.$$

Proof. Boundedness of $t^q|\log t|^2$ for q in a compact positive interval gives the expansion

$$I_{p+h}(\rho) = I_p(\rho) + hJ_p(\rho) + O(h^2)$$

uniformly over probability measures. Compare μ_n with any fixed $\nu \in \mathcal{M}_p$, use $I_p(\mu_n) \geq e(p)$, divide by $h_n > 0$, and pass to a weak limit. This gives $J_p(\mu) \leq J_p(\nu)$; a negative increment reverses the inequality. For the derivative formula from the right, comparison with a J_p-minimizer supplies the upper bound, while the expansion at μ_n and $I_p(\mu_n) \geq e(p)$ supply the lower bound. Compactness makes them agree. The left derivative is analogous. □

For any fixed N, the set $\mathcal{P}_{\le N}(X)$ of probability measures with at most N atoms is weakly compact: it is the image of the compact space $\Delta_{N-1} \times X^N$ under $(w, x) \mapsto \sum_i w_i \delta_{x_i}$. Therefore a uniformly bounded sequence of exact finite minimizers proves existence of a finite minimizing limit. Without a recovery argument, it does not prove that a prescribed co-minimizer is finite. Also, an unbounded sequence of support sizes need not have an infinite-support limit; atoms may coalesce or lose mass.

7.2. **Recovering a prescribed minimizer.** The following result addresses selection, but intentionally leaves support bounds as a separate question.

Theorem 7.2 (Target-centered recovery)**.** *Let X be compact metric and K_n, K continuous symmetric kernels with $\delta_n = \|K_n - K\|_\infty \to 0$. Fix any minimizer μ of I_K. There is a continuous positive semidefinite kernel H_μ such that, for every $\varepsilon_n > 0$ with $\varepsilon_n \to 0$ and $\delta_n/\varepsilon_n \to 0$, every choice of minimizers ν_n of $I_{K_n + \varepsilon_n H_\mu}$ satisfies $\nu_n \rightharpoonup \mu$.*

Proof. Choose a countable dense family (ϕ_j) in the unit ball of $C(X)$ and let

$$H(x, y) = \sum_{j \ge 1} 2^{-j} \phi_j(x) \phi_j(y).$$

The series converges uniformly and defines a continuous positive semidefinite kernel. The quantity

$$D_H(\rho, \mu)^2 = I_H(\rho - \mu) = \sum_{j \ge 1} 2^{-j} \left(\int \phi_j \, d(\rho - \mu) \right)^2$$

vanishes only when $\rho = \mu$. Center its feature map at μ:

$$H_\mu(x, y) = H(x, y) - \int H(x, z) \, d\mu(z) - \int H(z, y) \, d\mu(z) + I_H(\mu).$$

This is again positive semidefinite and satisfies $I_{H_\mu}(\rho) = D_H(\rho, \mu)^2$ for probability measures. If $e = I_K(\mu)$, comparison of ν_n with μ gives

$$e - \delta_n + \varepsilon_n D_H(\nu_n, \mu)^2 \le I_{K_n}(\nu_n) + \varepsilon_n D_H(\nu_n, \mu)^2 \le I_{K_n}(\mu) \le e + \delta_n.$$

Consequently

$$\varepsilon_n D_H(\nu_n, \mu)^2 \le 2\delta_n. \tag{7.2}$$

By compactness, continuity of H, and separation of measures by the ϕ_j, every cluster point is μ. □

Corollary 7.3. *Under the hypotheses of Theorem 7.2, if one can choose recovering minimizers with a common bound $\#\operatorname{supp} \nu_n \le N_\mu < \infty$, then $\#\operatorname{supp} \mu \le N_\mu$. If this hypothesis holds for every prescribed minimizer μ, then every minimizer of I_K is finitely supported.*

Proof. Apply weak compactness of $\mathcal{P}_{\le N_\mu}(X)$ to the recovery theorem. □

For the sphere-power family, one may take $K_n = K_{p_n}$ with rational $p_n \to p$ and, for instance, $\varepsilon_n = |p_n - p|^{1/2}$ when $p_n \ne p$. Equation (7.1) verifies the required ratio. However, Theorem 1.1 concerns the *unmodified* rational kernels, not the target-centered kernels in this construction. A uniform support bound for the latter remains an additional input. Likewise, Theorem 1.2 reduces irrational discreteness to a rank question, and Theorem 1.3 reduces robust local finiteness to all-contact nondegeneracy; neither missing hypothesis follows from finite approximation alone.

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